

SATHISH KUMAR

Full-Stack .NET Engineer | React | AI-Integrated Systems (RAG, LLM APIs) | 4+ Years Experience

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Portfolio: personal-portfolio-psi-eight-17.vercel.app | LinkedIn: [linkedin.com/in/\[your-username\]](https://linkedin.com/in/[your-username]) | GitHub: [github.com/\[your-username\]](https://github.com/[your-username])

PROFESSIONAL SUMMARY

Full-stack .NET engineer with 4+ years of experience designing and building scalable enterprise web applications using .NET Core, ASP.NET Core Web API, SQL Server, React, and Angular. Currently building AI-integrated features in production at Hitachi Energy — an LLM-powered workflow that converts natural-language customer requirements into validated engineering calculations and plain-language result summaries. Improved API and database performance by 25–30% in production systems through query optimization and clean architecture. Hands-on with LLM APIs, retrieval-augmented generation (RAG), prompt engineering, and agentic workflows.

TECHNICAL SKILLS

Backend: .NET Core, ASP.NET Core Web API, C#, Entity Framework Core, LINQ, REST APIs, Asynchronous Programming (async/await), Middleware, Clean Architecture, Repository & Unit-of-Work Patterns, Dependency Injection, JWT Authentication, Caching

AI / GenAI: LLM API Integration (OpenAI / Azure OpenAI), Function / Tool Calling, Structured Outputs, Prompt Engineering, Retrieval-Augmented Generation (RAG), Embeddings & Vector Search, Agentic Workflows, Machine Learning Fundamentals (ML, Deep Learning, NLP)

Database: SQL Server, T-SQL, Stored Procedures, Query Optimization, Indexing, Execution Plans, MongoDB

Frontend: React (Redux Toolkit, RTK Query), Angular (NgRx), JavaScript, TypeScript, HTML5, CSS3, Bootstrap

Cloud & DevOps: Azure DevOps, Docker, CI/CD Pipelines, Git, IIS; Python (FastAPI) for AI services

WORK EXPERIENCE

Software Engineer | Hitachi Energy, Chennai Feb 2025 – Present

- Develop an Instrument Transformer calculation platform for energy-sector customers using React and ASP.NET Core Web API, serving high-volume production workloads.
- Built an AI-assisted requirements workflow: an LLM interprets customer requirements described in natural language, auto-fills the engineering calculation form using structured outputs, invokes the calculation API, and generates a plain-language summary of results for the customer — reducing manual form entry and specification errors.
- Designed the LLM integration on the .NET backend with prompt engineering, function/tool calling, and validation guardrails to keep AI-generated inputs consistent with engineering constraints.
- Optimized SQL queries and stored procedures, improving data retrieval performance by up to 25% in production systems.
- Designed scalable backend services following clean architecture principles; built reusable React components that improved UI performance and development speed across teams.

Software Developer | Afaiez Semiconductor Pvt Ltd, Chennai Dec 2023 – Dec 2024

- Developed .NET Core Web API applications using repository and unit-of-work patterns for HRMS, CRM, and asset management products.
- Built HRMS modules for payroll, attendance, and employee management with role-based access control; automated workflows reduced manual HR effort by 30%.
- Developed CRM modules for customer tracking and service delivery, and asset management APIs handling large datasets with pagination and filtering.
- Improved UI responsiveness across products using Bootstrap, CSS Flexbox, and Grid.

Software Developer | Research Community, Chennai May 2022 – Dec 2023

- Built ASP.NET Core Web API applications with scalable architecture and repository pattern for improved maintainability.
- Developed dynamic UI tables with AJAX-based pagination, filtering, and sorting over large datasets.

- Integrated third-party Geocoding APIs for real-time location features, achieving ~95% accuracy in location data retrieval with response caching to cut API costs.
- Implemented email scheduling systems using SQL Server jobs and stored procedures.

KEY PROJECTS

RAG Document Chatbot (Full-Stack GenAI) — .NET 8 Web API, React, LLM API (Llama 3.3 via Groq)

- Built a full-stack retrieval-augmented generation (RAG) chatbot: users upload PDF/text documents and ask questions answered strictly from their content, with source citations and similarity scores shown for every answer.
- Implemented the complete RAG pipeline in .NET — text extraction, overlapping chunking, embeddings, and cosine-similarity vector search — with grounded prompting and low temperature to prevent hallucination.
- Designed provider-agnostic LLM and embedding abstractions (Groq / OpenAI swappable via configuration) with an offline mock mode for testing the pipeline without an API key. Source: [github.com/\[your-username\]/\[repo\]](https://github.com/[your-username]/[repo])

QueryPilot — Natural Language to Database Queries — .NET 8 Web API, React + Redux Toolkit, MongoDB

- Built an app that converts plain-English questions into validated MongoDB aggregation pipelines via LLM structured outputs, executes them read-only, and summarizes results in natural language.
- Implemented a security guard treating LLM output as untrusted input: collection whitelisting, forbidden-operator scanning (\$out, \$merge, \$where), and enforced result limits to block prompt-injection attacks. Source: [github.com/\[your-username\]/\[repo\]](https://github.com/[your-username]/[repo])

TriageBot — Multi-Agent Support Ticket Automation — .NET 8 Web API, React + RTK Query, MongoDB

- Built a three-agent pipeline — classifier (category/urgency), knowledge-base retriever, and reply drafter — with each agent's output persisted for full transparency in a review dashboard.
- Designed human-in-the-loop approval: AI-drafted replies grounded in KB citations are never sent without reviewer sign-off. Source: [github.com/\[your-username\]/\[repo\]](https://github.com/[your-username]/[repo])

EDUCATION

B.E. Mechanical Engineering

[Year of graduation]

RVS Padhmavathy College of Engineering and Technology, Chennai